

Seminar Presentation on

Wind Profile Estimation for K-Band Doppler Radar Based on Mutual Convolution Cost Function Method

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1. Introduction

1.1 Background

- Wind profiles are estimated from Doppler Radars via signal processing techniques.
- Radar echoes are often heavily corrupted by background noise (gets worse in cold climate)
- Traditional methods (like coherent accumulation) works well only when SNR is high
- Under low SNR conditions, multiple peaks are observed

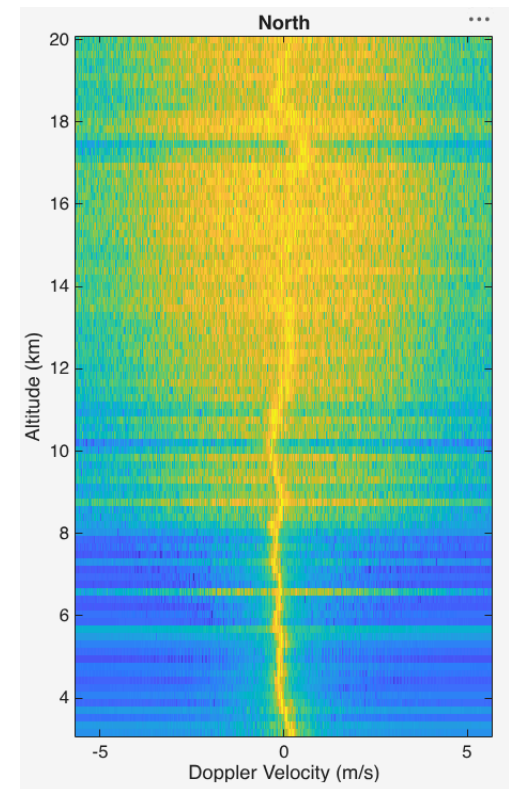


Figure 1: Doppler spectrum

1.2 Earlier Works

- **Clothiaux et al. [2]** proposed identifying spectral peaks by selecting the longest chain and the maximum sum of power spectra values as the principle
- **Anandan et al. [3]** proposed a method called adaptive moment estimation (AME) based on SNR and wind shear.
- **Sinha et al. [4], [5]** proposed the multiparameter cost function (MPCF) method for estimating the wind profile.
- **Li et al. [6]** proposed an improved moment estimation (IME) method for wind estimation.
- **Bhatta et al. [7]** proposed the Viterbi data association (VDA) method to track the wind profile.

In the field of turbulence detection, the traditional denoising method is coherent accumulation (CA)

- **Took and Mandic [8]** proposed the quaternion least mean square (QLMS) for wind vector evaluation based on CLMS.

1.2 Earlier Works

- **Pei and Ding [9]** proposed the fractional Fourier transform (FRFT) method to make the energy of unstable signals more concentrated to improve the filtering performance of adaptive filtering for unstable signals.
- **Chen et al. [10]** proposed the normalized leakage LMS (NL-LMS) to maximize the energy of the target signal and suppress clutter in the FRFT domain.
- **Shi et al. [11]** proposed shrinkage widely linear CLMS (SWL-CLMS), which significantly improved the convergence speed and reduced the steady-state misalignment.
- **Menguc and Acir [12]** proposed the augmented complex valued least mean kurtosis (ACLMK) filter, which utilizes augmented statistics to define the negative kurtosis of the complex error signal as a cost function.
- **Zhang et al. [13]** proposed a wide linear complex estimation input adaptive filter (WLC-EIAF), designed to achieve unbiased estimation in scenarios where input and output signals are affected by noise interference.

1.3 About Proposed Method

- Paper proposes a new method to estimate wind profiles from doppler radars by combining several methods including MCCF. [1]
- Works well under low SNR and multiple peaks conditions.
- The paper primarily focuses K-Band radar and near surface wind profile and on its potential application to efficient wind energy generation.

2. Experimental Setup

2. Experimental Setup

- Experiment equipment: K-band Vortex Radar (VORTRAD)
- Ground truth: LIDAR (usually has 99% accuracy in clear sky)
- At the end, correlation among LiDAR and Radar is obtained



Figure 2: Experiment Setup

2.1 Overview

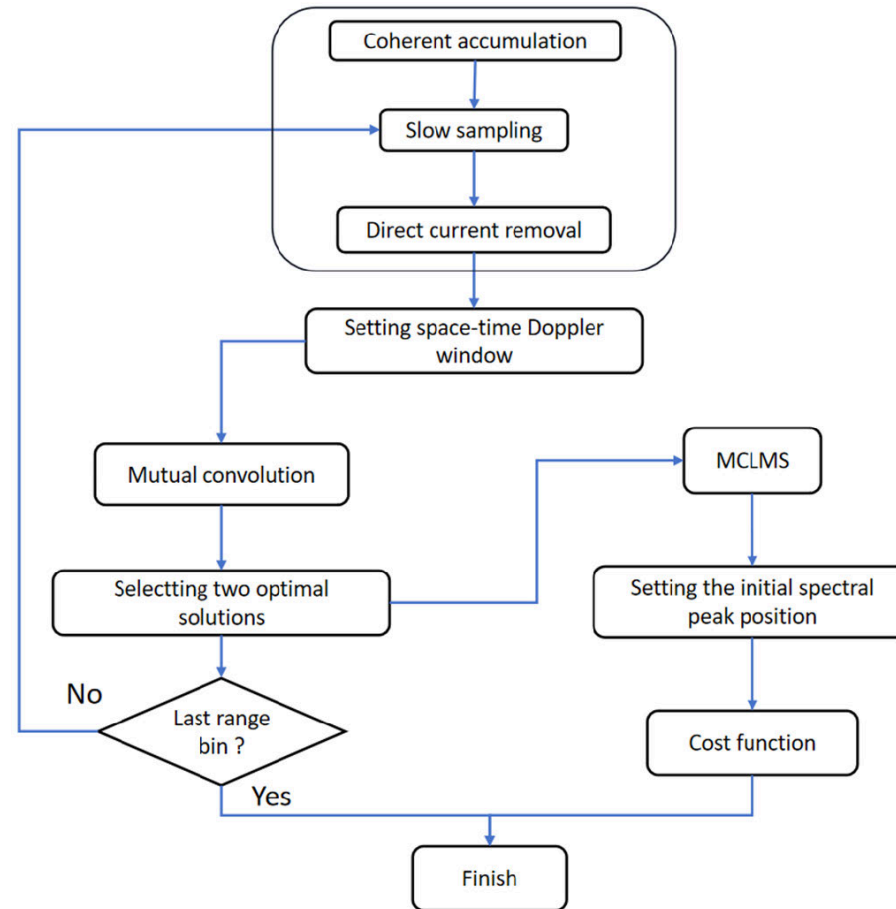


Figure 3: Flow diagram of method given by authors

3. Proposed Method

3.1 Signal Preprocessing

3.1.1 Coherent Accumulation

The CA adds and averages short segments of radar signals improving SNR
Result after N-order CA is represented as:

$$X = [x_{c1}^T, x_{c2}^T, \dots, x_{c(N-1)}^T, x_{cN}^T]$$

Where x_{ci} is result of i^{th} CA.

3.1.2 Slow Sampling

- On every Pulse Repetition Interval (PRI), a pulse is transmitted, radar receives echoes from all range bins (heights)
- For a single height, there will be 1 echo sample from every transmitted pulse.
- To obtain this, a row vector from X is used. This process is called *slow sampling* in the distance dimension

3.1 Signal Preprocessing

3.1.3 DC and Clutter Removal

- VORTRAD produces large DC power and ground clutter due to its zero-IF RF transceiver architecture.
- Here these are removed.
- The paper doesn't mention a particular clutter removal method

3.2 Setting Space-Time Doppler Window

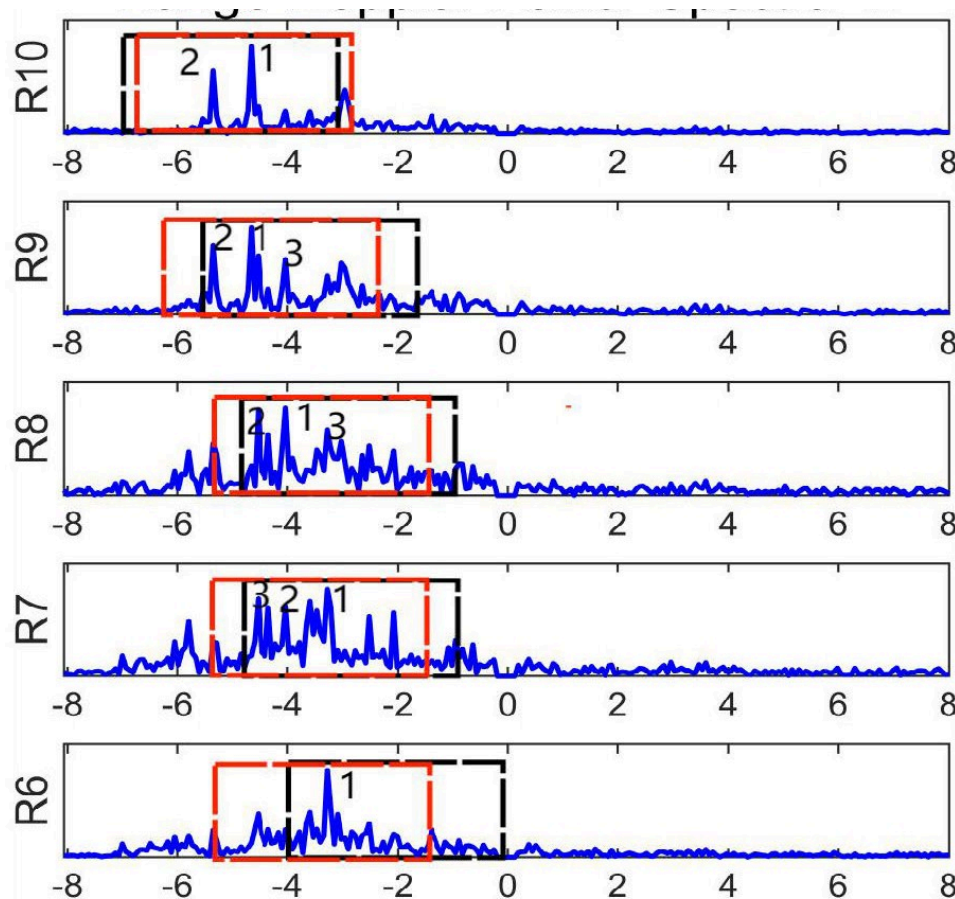


Figure 4: ST-W Window

- Idea is to restrict search region in both space and time (ST-W) for finding candidate peaks
- Assumes wind velocity changes smoothly with altitude (space) and between successive radar scans (time).
- In the figure
 - ▶ Red -> Space window
 - ▶ Black -> Time window

3.2 Setting Space-Time Doppler Window

Space window: $W_1 = (F_0 - \alpha d_f, F_0 + \alpha d_f)$

Time window: $W_2 = (T_0 - \alpha d_f, T_0 + \alpha d_f)$

$$\text{Space Time window: ST-W} = \begin{cases} W_1 \cap W_2 & \text{if } W_1 \cap W_2 \geq \alpha d_f \\ W_1 & \text{else} \end{cases}$$

Where

$$\text{wind shear } d_f = \frac{V_d}{d_v} = \frac{V_d}{f_{d\min} \lambda / L}$$

V_d = maximum value of wind shear

L = number of points in FFT.

α = window coefficient

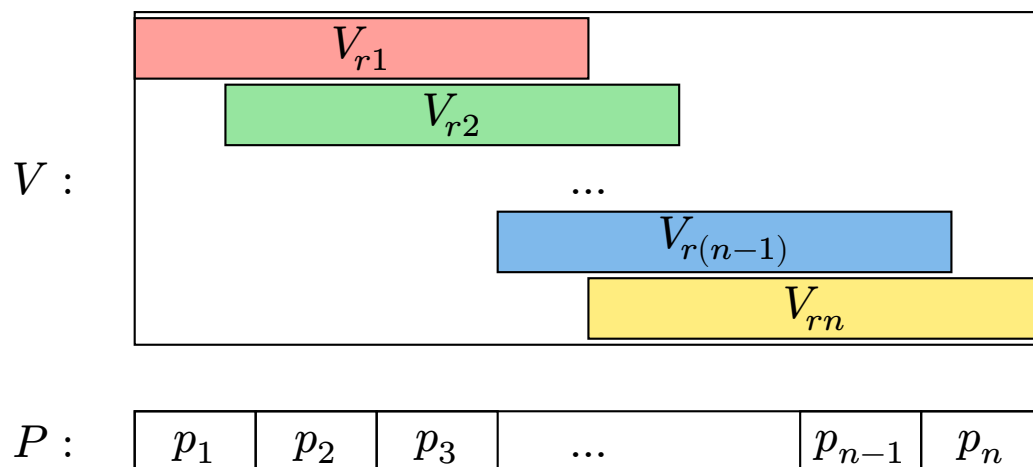
F_0 and T_0 : Space and time spectrum estimation
result of the previous range bin

3.3 Mutual Convolution

- Main idea is to use the temporal correlation of echoes to suppress uncorrelated noise.

Method:

1. Take a row vector $P = [p_1, p_2, \dots, p_N]$ of one range bin from X
2. Create new vectors from P with equal length (say half size) $V_{r1} = [p_1, p_2, \dots, p_{N/2}]$
3. From this a new vector V can be written as: $V = [V_{r1}, V_{r2}, \dots, V_{rn}]$ Where V_{ri} is segment of datas from different intervals in same range bin



3.3 Mutual Convolution

4. Mutual convolution is performed:

$$V_{C1} = V_{r2} * V_{r1}^{\text{HR}}$$

$$V_{C2} = V_{r3} * V_{r1}^{\text{HR}}$$

$$\vdots$$

$$V_{C(n-1)} = V_{rn} * V_{r1}^{\text{HR}}$$

$x^{\text{HR}} \rightarrow$ take conjugate and then reverse order

5. Extracting the middle of the vector (To “maintain the same time complexity and velocity resolution of calculation”):

3.3 Mutual Convolution

$$[V'_C] = \begin{bmatrix} V'_{C1} \\ V'_{C2} \\ \vdots \\ V'_{C(n-1)} \end{bmatrix} = \begin{bmatrix} V_{C1} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ V_{C2} \left(\frac{N}{4} : \frac{3N}{4} \right) \\ \vdots \\ V_{C(n-1)} \left(\frac{N}{4} : \frac{3N}{4} \right) \end{bmatrix}$$

6. Then maximum of power spectra of each spectrum is computed:

$$V'_{cf} = \left[\max \left\{ \left| \mathcal{F}([V'_c]) \right| \right\} \right]$$

7. We take a optimal solution X_r from convolution results based on maximum power V'_{cf} in . Also convolution results of adjacent delays are recorded as suboptimal solutions X'_r

3.4 MCLMS

- ~ An “adaptive filter”, pretrained.
- X_r and X'_r are taken input and desired signal of the filter.
- After the coefficients have been fully updated, filter can recognize stable signals and removes multiple strong clutter.

For finding Filter coefficients:

1. Calculate the normalized parameters of filter coefficients as:

$$u = 1 / \left(\sum_{j=1}^{N/2} |X_r(j)|^2 \right)$$

2. Construct “ X ” from the samples of X_r :

3.4 MCLMS

$$X = \begin{cases} \text{zeros}(1, k - j), X_r(1 : j) & , j < k \\ X_r(j - k + 1 : j) & , j \geq k \end{cases} \quad j \in \{1, 2, \dots, N/2\}$$

(k - length of “coefficent vector”)

3. Initialize filter coefficients as $W_1 = 0$. The coefficients are then updated using:

$$Y_j = W_1 X^T$$

$$e_j = X'_{rj} - Y_j$$

$$W'_1 = 0.98W_1 + ue_j X^H$$

4. Repeat above steps until all data are filtered, the coefficients have been fully updated

5. We will use this coefficent to obtain the filtering result Y_r

3.5 Generating the Cost Function

- To find out *candidate peaks*.

Steps

1. Calculate the average value corresponding to the power spectrum of the first range bin and pick out the prominent peak:

$$M_{1,1} = \max \left| \frac{1}{n-1} \sum_{i=1}^{n-1} \mathcal{F}(V'_{Ci}) \right|$$

Use this as starting point for ST–W

2. Compute doppler spectrum of each height bin

$$M_r = |\mathcal{F}(Y_r)|$$

3. In each ST–W, spectral peaks are sorted and up to 3 peaks are selected.

3.5 Generating the Cost Function

4. If $M_{r_{\max}}$ is spectral peak, then candidate peaks are:

$$M_{r,j} \geq 0.4M_{r_{\max}}$$

5. Cost value for each peak is:

$$V_{r+1,j} = 1 - \left| \frac{(M_{r+1} - M_r)d_v}{\alpha V_d} \right| \quad [d_v = \text{velocity resolution}]$$

6. MCCF cost function:

$$F_{r+1,j} = w_1 V_{r+1,j} + w_2 M_{r+1,j} \quad \text{and } w_1 + w_2 = 1$$

w_1, w_2 are constraint weights

7. The peak with the highest cost value is selected
8. Repeat this for all height bins

4. Results

4.1 Performance at Different Delay Times

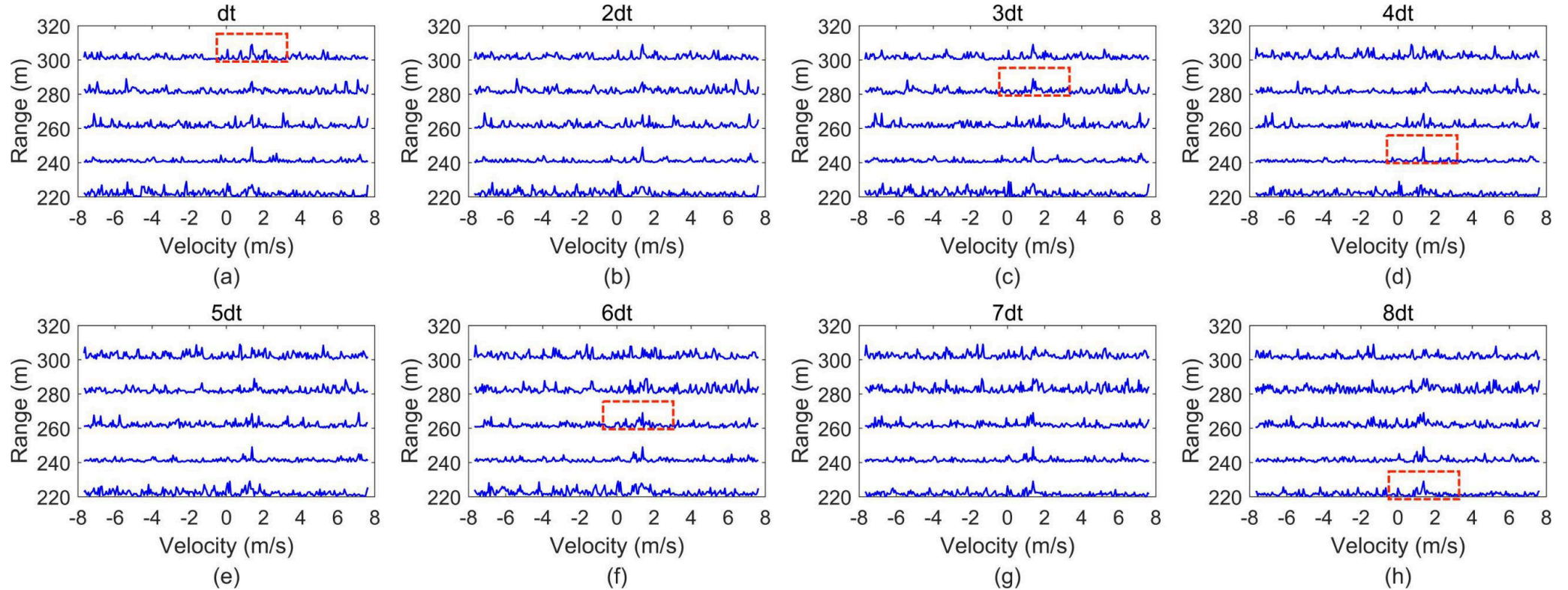


Fig: Optimal solutions are marked by red dotted box of different range bin after mutual convolution (a)–(h) Represent the filtering effect of dt – $8dt$ delay time respectively.

4.1 Performance at Different Delay Times

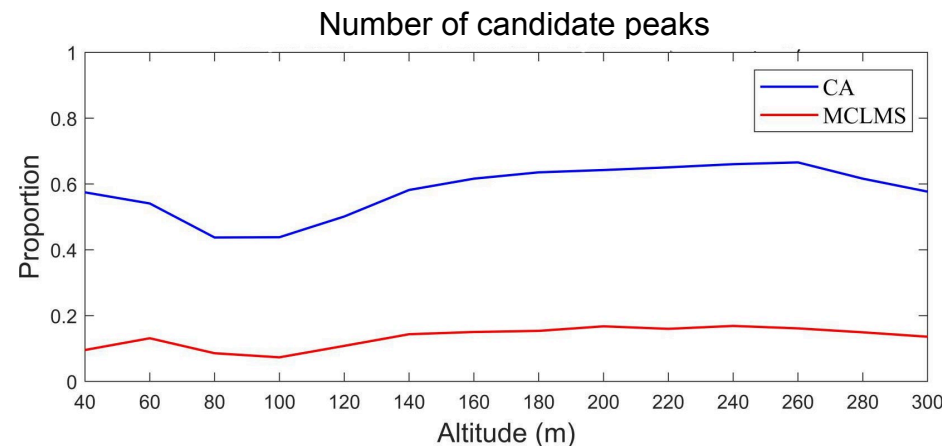
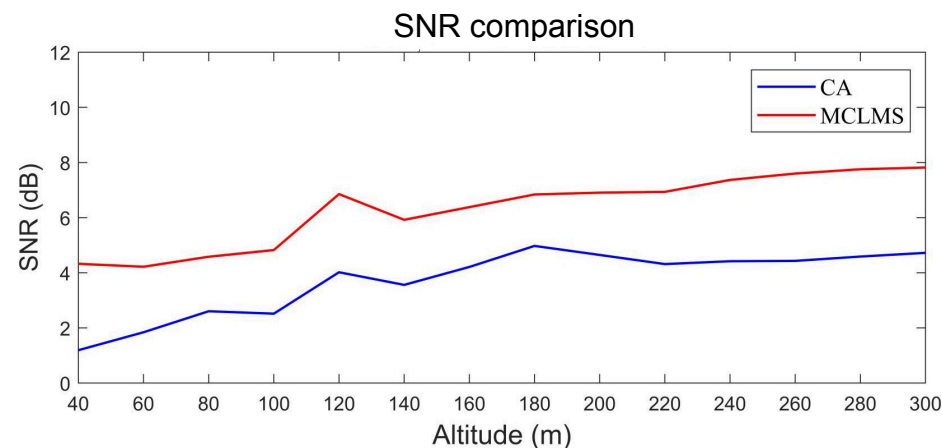
- As the delay time increases, the number of statistics gradually decreases
- Still not possible to determine a specific optimal delay time.

Delay	R1	R2	R3	R4	R5	R6	R7	R8
d_t	1543	1503	1568	1471	1486	1488	1498	1568
$2d_t$	444	404	430	420	398	397	406	403
$3d_t$	179	203	195	224	214	218	223	196
$4d_t$	128	129	91	119	114	127	122	126
$5d_t$	70	72	68	73	85	80	78	70
$6d_t$	39	63	58	69	75	71	66	35
$7d_t$	42	60	38	63	59	51	48	47
$8d_t$	55	66	52	61	69	68	59	55

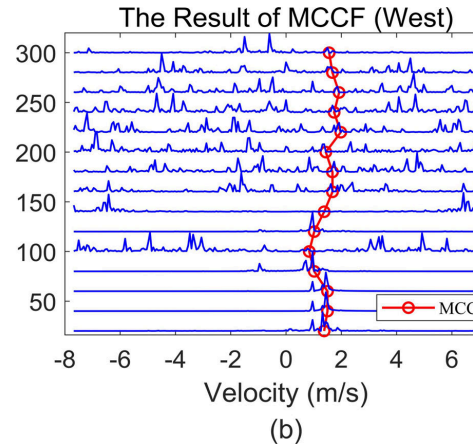
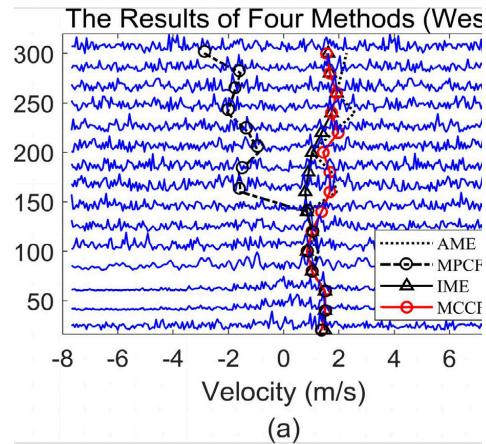
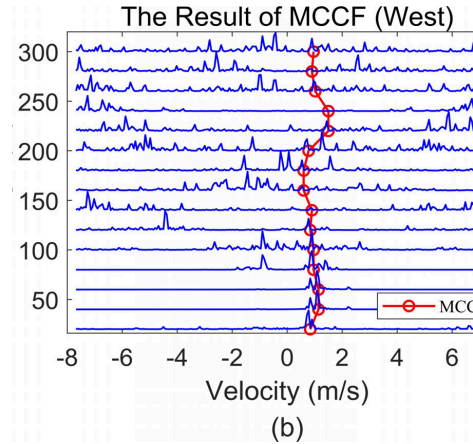
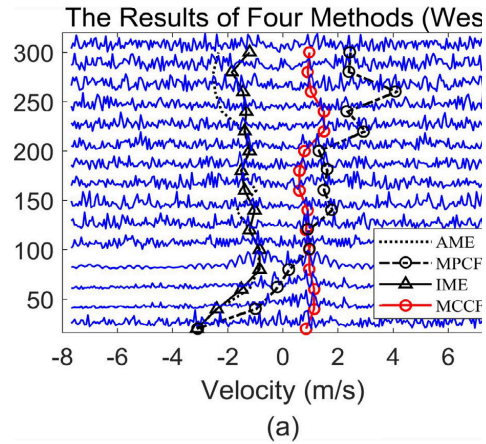
Table 1: Statistics of optimal delay time for mutual convolution

4.2 Comparison of Peak Number and SNR

- Average SNR for all bins plotted for CA and proposed MCLMS filters at different altitudes
- Proportion of spectra containing more than six candidate peaks



4.3 Wind Profile Estimation Results



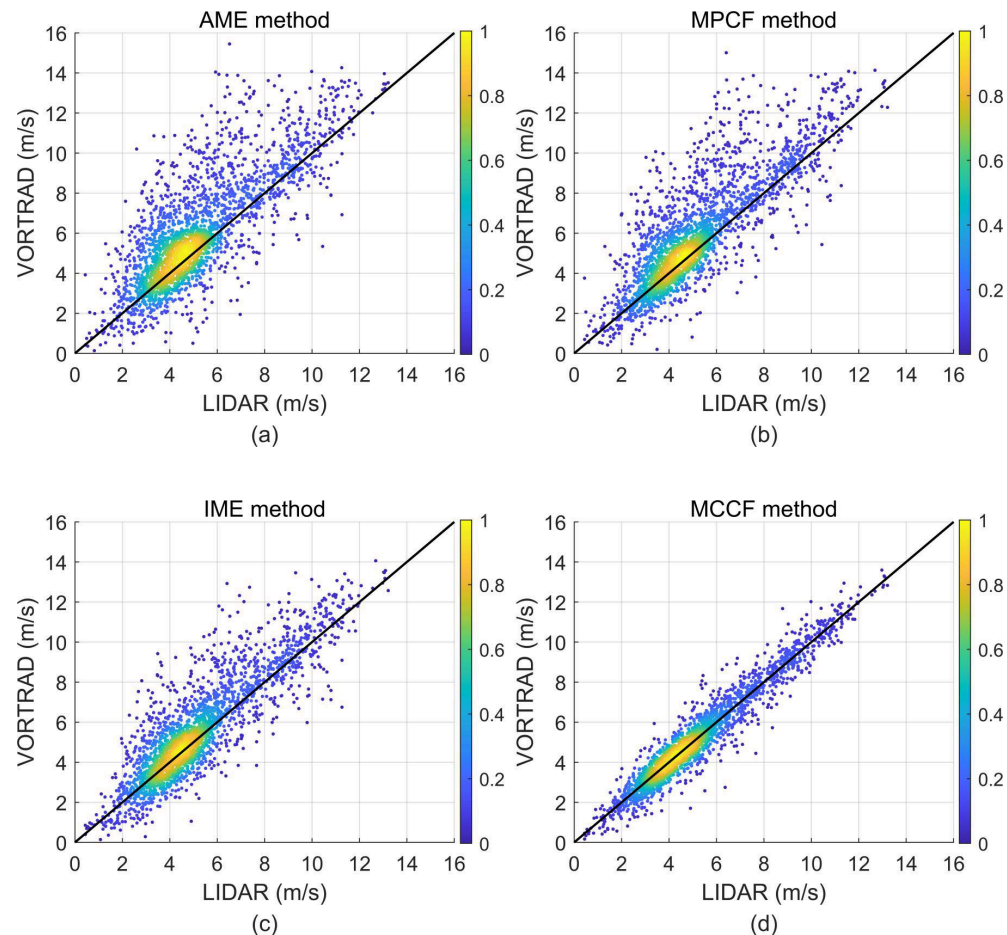
Comparison of four methods based on the data recorded on

- Dec 18, 2022 (15:19:32) (top row)
- Dec 18, 2022 (15:19:33) (bottom row)

4.4 Correlation with LiDAR observation

- IME shows some improvements over the AME and MPCF
- However MCCF aligns noticeably well with the LIDAR
- Correlation coefficients are summarized below:

Height(m)	AME	MPCF	IME	MCCF
40	0.8417	0.8368	0.8369	0.9576
80	0.8319	0.8350	0.8461	0.9609
120	0.8320	0.8431	0.8695	0.9625
160	0.8137	0.8353	0.8761	0.9641
200	0.7908	0.8276	0.8764	0.9647
240	0.7830	0.8202	0.8738	0.9650
280	0.7837	0.8219	0.8744	0.9651



5. Conclusion & Limitations

5. Conclusion & Limitations

- The proposed algorithm was applied on data collected by the VORTRAD K-band wind profiler and verified against LiDAR observation
- Method reduced false spectral peaks and improved the SNR ratio by approximately 2–4 dB
- Algorithm produces smoother and more continuous wind profiles
- Achieves highest correlation coefficient (0.96) with LiDAR observation.
- Authors note that the current method is validated only for clear-weather conditions and does not account for precipitation echoes such as rain or snow

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Thank You